

Electrochemistry & Applications

Complete Formula & Concept Revision Sheet

Last-day revision toolkit — all key formulas, reactions, mechanisms, and exam tips in one place

PROGRAM

B.Tech 1st Year

SEMESTER

Semester I

SUBJECT

Engg. Chemistry

UNIT

Unit II



Electrochemical Cell

- ▶ Device that converts **chemical energy → electrical energy**
- ▶ Consists of two half-cells (electrodes) connected by a salt bridge
- ▶ Involves spontaneous redox reactions

Electrolytic Cell

- ▶ Converts **electrical energy → chemical energy**
- ▶ Non-spontaneous reactions driven by external EMF
- ▶ Used in electroplating, electrolysis

Feature	Galvanic (Electrochemical)	Electrolytic Cell
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Reaction Type	Spontaneous ($\Delta G < 0$)	Non-spontaneous ($\Delta G > 0$)
Anode	Negative (-)	Positive (+)
Cathode	Positive (+)	Negative (-)
External Power	Not required	Required
Example	Daniel Cell, Batteries	Electroplating, Electrolysis

⚡ CELL NOTATION & EMF

Cell Representation (Daniel Cell):



Left = Anode (oxidation) | Double line = salt bridge | Right = Cathode (reduction)

EMF OF CELL

Also written as: $E_{\text{cell}} = E_{\text{reduction}}(+)-E_{\text{reduction}}(-)$

STANDARD EMF

Measured at 298K, 1 atm, 1M concentration

RELATIONSHIP WITH GIBBS ENERGY

n = moles of electrons transferred; F = 96485 C/mol (Faraday's constant)



NERNST EQUATION

GENERAL (THERMODYNAMIC) FORM

$R = 8.314 \text{ J/mol}\cdot\text{K}$ | $T = \text{Temperature (K)}$ | $Q = \text{Reaction quotient}$

LOGARITHMIC FORM (AT 298 K)

$0.0592 = (2.303 \times R \times 298) / F \approx 0.0592 \text{ V}$ (use 0.059 in MCQs)

REACTION QUOTIENT Q (FOR $AA + BB \rightarrow CC + DD$)

At equilibrium: $Q = K_{\text{eq}}$ and $E_{\text{cell}} = 0$

✓ Key Points:

- ▶ $E^{\circ}_{\text{cell}} > 0 \rightarrow \text{spontaneous reaction}$
- ▶ $E^{\circ}_{\text{cell}} < 0 \rightarrow \text{non-spontaneous}$
- ▶ Higher concentration at anode $\rightarrow$ lower EMF

⚠ Common Mistakes:

- ▶ Using log instead of ln ($\times 2.303$)
- ▶ Wrong sign: $E_{\text{anode}} - E_{\text{cathode}}$
- ▶ Forgetting to substitute T in Kelvin



PRIMARY CELL — ZINC-AIR BATTERY

Construction

- ▶ **Anode:** Zinc (Zn) powder
- ▶ **Cathode:** Porous carbon + air (O₂)
- ▶ **Electrolyte:** KOH (alkaline)
- ▶ **Separator:** Ion-exchange membrane

Key Facts

- ▶ High energy density (~1350 Wh/kg)
- ▶ Voltage: ~1.65 V
- ▶ Non-rechargeable (primary)
- ▶ Uses atmospheric oxygen as cathode reactant

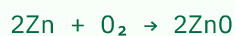
Anode (oxidation)



Cathode (reduction)



Overall Reaction



SECONDARY CELL — NICKEL-CADMIUM (NI-CD)

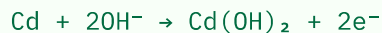
Construction

- ▶ **Anode:** Cadmium (Cd)
- ▶ **Cathode:** Nickel oxyhydroxide [NiO(OH)]
- ▶ **Electrolyte:** KOH (30%)
- ▶ Voltage: ~1.2 V per cell

Applications

- ▶ Portable electronics, power tools
- ▶ Emergency lighting
- ▶ Aircraft & satellite systems
- ▶ Rechargeable (500–1000 cycles)

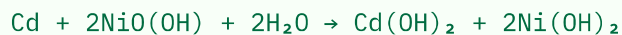
Discharge — Anode (oxidation)



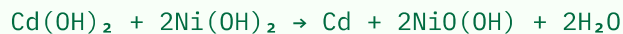
Discharge — Cathode (reduction)



Overall Discharge



Charging (reverse of discharge)



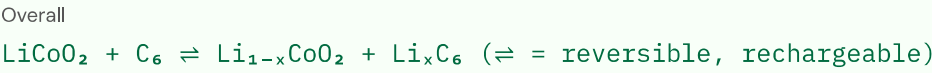
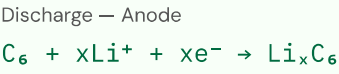
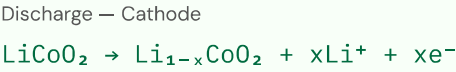
SECONDARY CELL — LITHIUM-ION BATTERY

Construction

- **Anode:** Graphite (intercalated Li)
- **Cathode:** LiCoO₂ / LiMnO₂
- **Electrolyte:** LiPF₆ in organic solvent
- Voltage: ~3.6 V per cell

Advantages

- High energy density (~150–200 Wh/kg)
- Low self-discharge (<5%/month)
- No memory effect
- Long cycle life (500–2000 cycles)



BATTERY COMPARISON TABLE

Feature	Zinc-Air	Ni-Cd	Li-Ion
Type	Primary	Secondary	Secondary
Voltage	1.65 V	1.2 V	3.6 V
Anode	Zinc	Cadmium	Graphite/Li
Cathode	O ₂ (air)	NiO(OH)	LiCoO ₂
Electrolyte	KOH	KOH	LiPF ₆ /organic
Energy Density	Very High	Moderate	High
Rechargeable?	No	Yes	Yes
Memory Effect	N/A	Yes	No
Applications	Hearing aids, watches	Power tools, aviation	Phones, EVs, laptops



What is a Fuel Cell?

- ▶ Electrochemical device that **continuously converts** chemical energy (from fuel) to electrical energy
- ▶ Operates as long as fuel and oxidant are supplied
- ▶ No combustion — directly produces electricity
- ▶ Efficiency: 40–70% (vs ~35% for thermal plants)

Fuel Cell vs Battery

Feature	Fuel Cell	Battery
Fuel	External (continuous)	Internal (stored)
Recharging	Not needed	Needed
Reactant	H ₂ , O ₂ , hydrocarbons	Electrodes
Lifetime	Very long	Limited cycles

 HYDROGEN-OXYGEN FUEL CELL

Construction

- ▶ **Anode:** Porous carbon + platinum catalyst
- ▶ **Cathode:** Porous carbon + platinum catalyst
- ▶ **Electrolyte:** KOH (alkaline type) or H₃PO₄ (acid type)
- ▶ Temperature: 60–200°C

Working Principle

- ▶ H₂ supplied at anode → oxidized
- ▶ O₂ supplied at cathode → reduced
- ▶ Ions travel through electrolyte
- ▶ Electrons flow through external circuit → current
- ▶ Byproduct: only water (clean!)

Anode (oxidation) — Alkaline



Cathode (reduction) — Alkaline



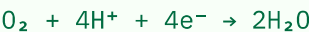
Overall Reaction



Anode (oxidation) — Acidic



Cathode (reduction) — Acidic



THEORETICAL EFFICIENCY OF H₂–O₂ FUEL CELL

✓ Advantages:

- Zero emission (only water produced)
- High efficiency compared to heat engines
- Quiet operation, no moving parts
- Continuous operation (no recharge)
- Scalable power output

Applications

- Space vehicles (NASA Apollo)
- Fuel cell electric vehicles (FCEVs)
- Submarines & military
- Portable power generation
- Backup power for hospitals



Definition

Corrosion is the **deterioration or degradation** of a metal or alloy due to its reaction with the environment (chemical or electrochemical), resulting in loss of material and properties.

Importance of Studying Corrosion

- Causes ~3–4% of GDP loss worldwide
- Structural failures in bridges, pipelines
- Safety hazards in nuclear plants
- Leads to development of protective coatings

ELECTROCHEMICAL THEORY OF CORROSION

Principle: Corrosion occurs due to formation of local galvanic cells on the metal surface where anodic and cathodic regions exist simultaneously.

Anodic Reaction (Metal Dissolution / Oxidation)



Cathodic Reaction (in neutral/alkaline, aerated solution)



Cathodic Reaction (in acidic solution)



Rust Formation (Iron)



TYPES OF CORROSION

1. Differential Aeration Corrosion

- Caused by **unequal oxygen concentration** at different parts of metal
- Region with **less O_2** → becomes **anode** → corrodes
- Region with **more O_2** → becomes **cathode**

Examples:

- Pitting corrosion under debris
- Iron pipe buried in soil at different depths
- Waterline corrosion on ship hull

2. Galvanic Corrosion

- Occurs when **two dissimilar metals** are in electrical contact in an electrolyte
- **More active metal** (higher in galvanic series) → acts as **anode** → corrodes
- Less active metal → cathode → protected
- Rate increases with larger potential difference

Examples:

- Zinc–Copper couple in seawater (Zn corrodes)
- Steel rivets in copper sheet
- Iron pipe connected to brass fitting

3. Dry (Chemical) Corrosion

- Occurs in absence of moisture/electrolyte
- Direct chemical reaction between metal and gas (O₂, SO₂, H₂S, etc.)
- Common at **high temperatures**
- Forms metal oxide layer on surface

Metal Oxide Formation:

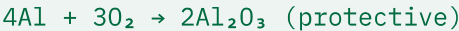
General



Iron



Aluminium



Feature	Differential Aeration	Galvanic	Dry/Chemical
Cause	O ₂ concentration difference	Dissimilar metals contact	Gas-metal reaction
Medium	Electrolyte required	Electrolyte required	No electrolyte needed
Temperature	Normal	Normal	Elevated temperature
Anode	Low O ₂ region	Active metal	Whole metal surface
Example	Pitting, waterline	Zn-Cu, Fe-Cu joints	Fe ₃ O ₄ , Al ₂ O ₃ formation



PILLING-BEDWORTH RATIO (PBR)

PBR FORMULA

$$PBR = \frac{M}{n \cdot \rho \cdot d}$$

M = Molar mass; ρ = Density; n = moles of metal in oxide formula

PBR < 1

- Oxide layer is **porous/cracked**
- Does NOT protect metal
- Corrosion continues
- Example: Alkali metals (Na, K, Ca)

PBR ≈ 1 (1–2)

- Oxide layer is **compact & adherent**
- Forms a **protective coating**
- Prevents further corrosion
- Example: Al₂O₃ (PBR = 1.28)

PBR > 2

- Oxide layer under **compressive stress**
- Flakes off / spalls
- Not protective
- Example: Fe₃O₄ (PBR = 2.1)

Metal	Oxide	PBR Value	Nature
Sodium (Na)	Na ₂ O	0.57	Non-protective
Magnesium (Mg)	MgO	0.81	Non-protective
Aluminium (Al)	Al ₂ O ₃	1.28	Protective
Chromium (Cr)	Cr ₂ O ₃	2.02	Protective
Iron (Fe)	Fe ₃ O ₄	2.10	Borderline/Spalls
Tungsten (W)	WO ₃	3.40	Non-protective

 FACTORS AFFECTING CORROSION RATE

Nature of Metal

- ▶ **Position in EMF series:** More active (anodic) metals corrode faster
- ▶ **Purity:** Impurities create galvanic couples → accelerate corrosion
- ▶ **Physical state:** Stressed/strained regions corrode faster
- ▶ **Passivation:** Al, Cr, Ti form protective oxide layers

Temperature

- ▶ Higher temp → higher diffusion rate → faster corrosion
- ▶ Dissolved O₂ decreases with temperature (counteracts)
- ▶ Kinetic factor dominates at high temperatures

Environment & Moisture

- ▶ **Humidity:** Critical humidity (60–65%) required for corrosion onset
- ▶ **Electrolytes:** Presence of salt (NaCl) accelerates corrosion
- ▶ **pH:** Acidic conditions → faster corrosion
- ▶ **O₂ concentration:** Higher O₂ → faster cathodic reaction

Impurities & Alloy Composition

- ▶ Impurities act as cathodes → increase local current
- ▶ Grain boundaries are more susceptible
- ▶ Alloying with Cr, Ni improves corrosion resistance

CATHODIC PROTECTION

1. Sacrificial Anode Method

- ▶ A more **active metal** (Zn, Mg, Al) is connected to the metal to be protected
- ▶ Active metal acts as anode → sacrificially corrodes
- ▶ Protected metal acts as cathode → protected
- ▶ Anode replaced periodically

Sacrificial Anode Reaction

$Mg \rightarrow Mg^{2+} + 2e^{-}$ (Mg corrodes, Fe protected)

Applications:

- ▶ Underground pipelines
- ▶ Ship hulls (Zn blocks)
- ▶ Water heater tanks

2. Impressed Current Method

- ▶ External DC current applied to make protected metal a **cathode**
- ▶ Inert anode used (graphite, platinum)
- ▶ Current drives away corrosive reactions
- ▶ Requires continuous power supply

Applications:

- ▶ Long-distance gas/oil pipelines
- ▶ Offshore platforms
- ▶ Bridges, harbor structures
- ▶ Reinforced concrete structures

ANODIC PROTECTION

- ▶ External anodic current applied to promote formation of a **stable passive oxide film**
- ▶ Metal enters passive region → corrosion rate drops drastically
- ▶ Suitable only for **passive metals** (Fe, Ni, Cr, stainless steel)
- ▶ Used in chemical industry (sulfuric acid storage, fertilizer plants)

Feature	Cathodic (Sacrificial)	Cathodic (Impressed)	Anodic
Mechanism	Active metal sacrificed	External DC current	Passive film formation
Power Required	No	Yes	Yes
Anode Material	Zn, Mg, Al	Graphite, Pt	Protected metal
Maintenance	Anode replacement	Power monitoring	Current control
Applications	Ships, pipelines	Buried structures	Acid storage tanks

OTHER PREVENTION METHODS

Surface Coatings

- ▶ **Metallic coatings:** Galvanizing (Zn on Fe), Tinning, Chromium plating
- ▶ **Chemical conversion:** Anodizing, phosphating, chromating
- ▶ **Organic coatings:** Paints, lacquers, varnishes, enamels

Alloying & Design

- ▶ Stainless steel (Fe + 12% Cr) → passive oxide layer
- ▶ Avoid dissimilar metal contacts
- ▶ Use corrosion inhibitors in electrolyte
- ▶ Design to avoid water retention



ELECTROPLATING

Principle

- Uses **electrolytic cell** principle
- Object to be plated = **cathode**
- Metal to be deposited = **anode**
- Electrolyte = salt solution of plating metal
- External DC current drives deposition

Process Steps

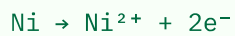
- **Step 1:** Surface preparation (cleaning, pickling)
- **Step 2:** Strike plating (thin flash layer)
- **Step 3:** Main plating bath
- **Step 4:** Post-treatment (rinsing, drying)

Nickel Plating

Cathode (Object)

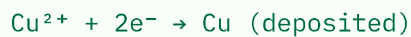


Anode (Ni metal)

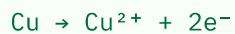
**Electrolyte:** $\text{NiSO}_4 + \text{NiCl}_2 + \text{H}_3\text{BO}_3$ (Watts Bath) | **pH:** 3.5–4.5 | **Temp:** 45–65°C**Uses:** Decorative, hardening surfaces, corrosion resistance for automotive parts

Copper Plating

Cathode



Anode (Copper metal)

**Electrolyte:** $\text{CuSO}_4 + \text{H}_2\text{SO}_4$ (acid bath) | **Uses:** PCBs, jewelry, base for other platings

FARADAY'S LAW OF ELECTROLYSIS

 $m = \text{mass deposited (g)} \mid M = \text{molar mass} \mid I = \text{current (A)} \mid t = \text{time (s)} \mid n = \text{valency} \mid F = 96485 \text{ C/mol}$

ELECTROLESS PLATING

Definition & Principle

- No external electric current required
- Deposition by autocatalytic chemical reduction
- Reducing agent in bath provides electrons
- Process is self-sustaining once initiated
- Common reducing agents: NaH₂PO₂ (hypophosphite), formaldehyde, hydrazine

Mechanism (Electroless Ni)

Nickel deposition



Reducing agent (hypophosphite)



Overall



Feature	Electroplating	Electroless Plating
Power Supply	External DC required	Not required
Reduction Agent	Electrons (external)	Chemical reducing agent
Substrate	Must be conducting	Any surface (metals, plastics)
Uniformity	Non-uniform on complex shapes	Highly uniform thickness
Cost	Lower (simpler setup)	Higher (complex chemistry)
Applications	Jewelry, automotive, PCBs	PCBs, aerospace, plastic plating
Adhesion	Good	Excellent



Q What is EMF of a cell?

A Potential difference between the two electrodes; driving force for current flow in the circuit. Unit: Volt (V)

Q What is the standard hydrogen electrode (SHE) potential?

A $E^\circ = 0.00 \text{ V}$ (by convention); used as reference for all standard electrode potentials

Q What happens at anode in galvanic cell?

A Oxidation occurs. Anode is the negative terminal. Metal loses electrons: $M \rightarrow M^{n+} + ne^-$

Q Byproduct of H_2 - O_2 fuel cell?

A Only water (H_2O). Makes it zero-emission. Overall: $2H_2 + O_2 \rightarrow 2H_2O$

Q Faraday's constant value?

A $F = 96485 \text{ C/mol} \approx 96500 \text{ C/mol}$. Charge of one mole of electrons

Q Li-ion battery cathode material?

A $LiCoO_2$ (lithium cobalt oxide). Anode: graphite. Electrolyte: $LiPF_6$ in organic solvent

Q Sacrificial anode method uses?

A More active metals (Zn, Mg, Al) connected to protected metal; they corrode preferentially as anode

Q Electroless plating requires?

A No external current; uses chemical reducing agent (NaH_2PO_2) for autocatalytic metal deposition

Q Salt bridge function?

A Maintains electrical neutrality in both half-cells by allowing ion transfer while preventing direct mixing

Q Zinc-air battery electrolyte?

A KOH (potassium hydroxide) – alkaline electrolyte. Uses atmospheric O_2 as cathode reactant

Q Nernst equation is used for?

A Calculating cell potential under non-standard conditions (variable concentration, temperature)

Q PBR significance?

A Indicates protective nature of metal oxide layer. PBR 1–2 = protective; <1 or >2 = non-protective

Q What happens at cathode in galvanic cell?

A Reduction occurs. Cathode is positive terminal. Ions gain electrons: $M^{n+} + ne^- \rightarrow M$

Q What is galvanic corrosion?

A Corrosion when two dissimilar metals contact in electrolyte; more active metal corrodes (becomes anode)

Q What is differential aeration corrosion?

A Corrosion due to unequal O_2 concentration; region with less O_2 becomes anode and corrodes

Q Why is Al corrosion resistant?

A Al_2O_3 has PBR = 1.28 (protective range). Dense, adherent oxide layer prevents further attack

Q What makes Ni-Cd battery rechargeable?

A Reversible electrochemical reactions: $Cd/Cd(OH)_2$ and $Ni(OH)_2/NiO(OH)$ redox couple

Q ΔG and E cell relation?

A $\Delta G^\circ = -nFE^\circ_{\text{cell}}$. Negative ΔG = spontaneous; positive E°_{cell} = spontaneous reaction

Q Dry corrosion mechanism?

A Direct chemical reaction between metal and gas (O_2 , SO_2 , Cl_2) without moisture; common at high temperatures

Q What is passivation?

A Formation of thin, stable, adherent oxide film on metal surface that inhibits further corrosion (e.g., Al, Cr, stainless steel)

Q Nernst equation constant at 298K?

A $0.0592/n$ (using $\log_{10}$). Derived from $(2.303RT/F)$ where $R=8.314$, $T=298$, $F=96485$

Q Fuel cell vs battery key difference?

A Fuel cell uses continuous external fuel supply; battery stores chemical energy internally and needs recharging

Q In electroplating, which electrode is the object?

A Cathode (negative terminal). Metal ions deposit on it. Anode is the plating metal source

Q Impressed current method uses?

A External DC current to make protected structure cathodic; inert anode (graphite/Pt); used for buried pipelines

Q Rust formula?

A $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (hydrated ferric oxide). Formed by: $4\text{Fe} + 3\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$

Q Critical humidity for corrosion?

A ~60–65% relative humidity. Below this, insufficient moisture for electrolyte film formation on metal surface

**⚠ NERNST EQUATION MISTAKES**

✗ Wrong: Using log without $\times 2.303$ factor
✓ Correct: $E = E^\circ - (0.0592/n) \log Q$ at 298K

✗ Wrong: $E_{\text{cell}} = E^\circ_{\text{anode}} - E^\circ_{\text{cathode}}$
✓ Correct: $E_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$

✗ Wrong: Putting T in $^\circ\text{C}$ in Nernst equation
✓ Correct: T must be in Kelvin ($K = ^\circ\text{C} + 273$)

✗ Wrong: Flipping Q (products/reactants reversed)
✓ Correct: $Q = [\text{Products}] / [\text{Reactants}]$ (follow stoichiometry)

⚠ CELL & EMF MISTAKES

✗ Wrong: Anode = positive in galvanic cell
✓ Correct: Anode = negative (-) in galvanic cell; cathode = positive (+)

✗ Wrong: Anode = negative in electrolytic cell
✓ Correct: Anode = positive (+) in electrolytic cell

✗ Wrong: Omitting units for E (cell potential)
✓ Correct: Always express in Volts (V)

⚠ BATTERY & FUEL CELL MISTAKES

✗ Wrong: Confusing Ni-Cd charging/discharging reactions
✓ Correct: Discharge: Cd oxidizes; Charging: $\text{Cd}(\text{OH})_2 \rightarrow \text{Cd}$ (reverse)

✗ Wrong: Saying fuel cells need recharging like batteries
✓ Correct: Fuel cells operate continuously as long as fuel is supplied

✗ Wrong: Fuel cell byproduct = CO_2
✓ Correct: $\text{H}_2\text{-O}_2$ fuel cell produces ONLY water (zero emission)

⚠ CORROSION MISTAKES

✗ Wrong: Higher O_2 region = anode (corrodes)
✓ Correct: Lower O_2 region = anode (corrodes) in differential aeration

✗ Wrong: PBR = 1.5 means non-protective
✓ Correct: PBR between 1 and 2 is protective (compact oxide)

✗ Wrong: Any PBR > 1 = protective
✓ Correct: PBR > 2 = non-protective (spalling due to stress)

✗ Wrong: Zinc coat on iron — Zn is cathode
✓ Correct: Zn is anode (more active); sacrificially protects Fe even when coating is scratched

💡 SCORING TIPS**✓ High-Yield Topics:**

- ▶ Nernst equation numerical problems (5 marks)
- ▶ Battery reaction equations (write balanced half-reactions)
- ▶ PBR calculation and interpretation (2–3 marks)
- ▶ Corrosion mechanism with diagram description
- ▶ Electroplating Faraday's law numericals

✓ Remember These Numbers:

- ▶ $F = 96485 \text{ C/mol} \approx 96500 \text{ C/mol}$
- ▶ $R = 8.314 \text{ J/mol}\cdot\text{K}$
- ▶ $0.0592 = (2.303 \times 8.314 \times 298) / 96485$
- ▶ Li-ion voltage: $\sim 3.6 \text{ V}$ | Ni-Cd: $\sim 1.2 \text{ V}$ | Zn-Air: $\sim 1.65 \text{ V}$
- ▶ PBR of $\text{Al}_2\text{O}_3 = 1.28$ (protective example)

 CORE FORMULAS

EMF

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

GIBBS ENERGY

$$\Delta G = -nFE_{\text{cell}}$$

NERNST (GENERAL)

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q$$

NERNST (AT 298K)

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592}{n} \log Q$$

PBR

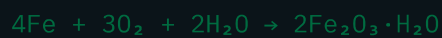
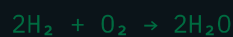
$$\frac{P_{\text{H}_2}}{P_{\text{H}_2}^{\circ}} = \frac{P_{\text{O}_2}}{P_{\text{O}_2}^{\circ}} = \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2\text{O}}^{\circ}}$$

FARADAY'S LAW

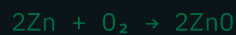
$$m = \frac{M}{zF} Q$$

 KEY REACTIONS

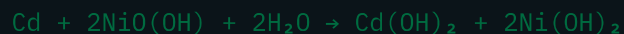
Rust

 H_2 - O_2 Fuel Cell

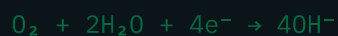
Zn-Air (Overall)



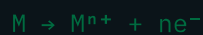
Ni-Cd Discharge



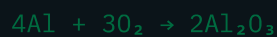
Corrosion (Cathodic)



Corrosion (Anodic)



Al Passivation



📌 KEY DEFINITIONS

- ▶ **EMF:** Potential difference driving current flow
- ▶ **Standard electrode potential:** Potential at 1M, 298K, 1 atm vs SHE
- ▶ **Corrosion:** Metal degradation by environment (electrochemical/chemical)
- ▶ **PBR:** Ratio of oxide volume to metal volume consumed
- ▶ **Passivation:** Protective stable oxide film on metal surface
- ▶ **Galvanic series:** Activity order of metals; guides corrosion prediction
- ▶ **Salt bridge:** Electrolytic connection maintaining charge neutrality
- ▶ **Sacrificial anode:** Active metal protecting cathode by corroding itself
- ▶ **Fuel cell:** Electrochemical device converting fuel energy to electricity continuously

⚡ QUICK RECALL TABLE

Concept	Keyword
Anode (Galvanic)	Oxidation, -ve
Cathode (Galvanic)	Reduction, +ve
Anode (Electrolytic)	Oxidation, +ve
PBR < 1	Porous, non-protective
PBR 1–2	Compact, protective
PBR > 2	Stress, spalling
Low ΔG zone	Anode → corrodes (GAL)
Active metal	Anode → corrodes (GC)
Galvanizing	Zn anode, sacrificial
Li-ion anode	Graphite (Li intercalated)

🏆 LAST-MINUTE MEMORY ANCHORS

OXIDATION

anode, loss of e⁻, OIL

OIL RIG: Oxidation Is Loss, Reduction Is Gain

NERNST TIP

Q has products on top

Q has products on top, reactants below

CORROSION

Low ΔG → Anode → Corrodes

Active metal → Anode in galvanic pair